TECHNICAL & OPERATIONAL SPECIFICATION

WhatsApp Chat Assistant

Bidirectional Conversational Ledger & Real-Time Financial Synchronization Architecture

Platform: **Nirvul360 / Gross Management**
Version: **2.0 Production**
Date: **November 2026**
Status: **Active Specification**

WhatsApp Chat Assistant — Implementation & Architecture Guide

This document is the comprehensive technical and operational specification for the **WhatsApp Chat Assistant** in the Nirvul360 / Gross Management platform.

It details how clients, merchants, and operations staff can submit miscellaneous expenses, courier fees, operational charges, and order status updates over WhatsApp, which are parsed, verified via a 2-step confirmation workflow, posted into the double-entry accounting ledger, and reflected in real time on the web dashboard.

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1. Executive Summary & Core Objectives

The Business Problem

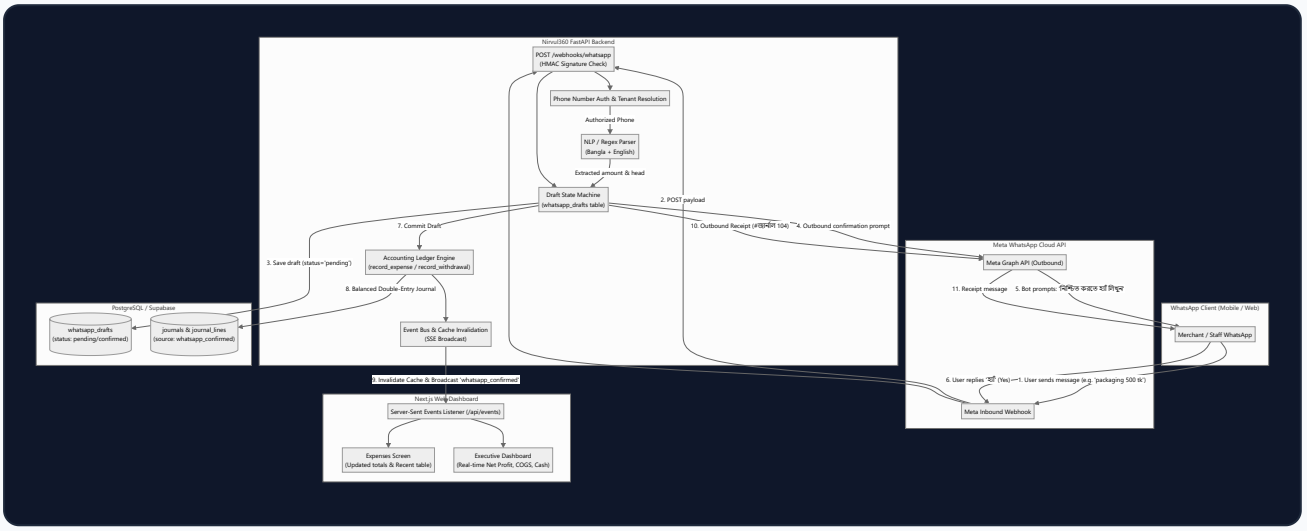
In Bangladeshi e-commerce and retail operations: - Up to 40% of petty cash expenses (packaging tapes, delivery bags, spot courier tips, transit rickshaw fare, tea/snacks) are paid in cash on the floor and never get logged into accounting systems. - Reconciling courier charges and consignment statuses (e.g. delivered, returned) often lags behind physical operations by several days. - Asking merchants or floor managers to open a complex web dashboard every time they spend ৳150 creates friction and compliance drop-off.

The Solution: WhatsApp Conversational Ledger

- **Zero-Friction Ingestion**: Users send casual messages in Bangla, English, or Banglish (e.g. `packaging 500 tk`, `কুরিয়ার চার্জ ১৫০ টাকা`, `order #1894 delivered`).
- **Strict Anti-Mistake Protection (2-Step Confirmation)**: Messages are never posted directly into accounting books. They are drafted and presented back to the user with exact amounts, heads, and dates. Posting occurs **only** when the user affirms (`হ্যাঁ` / `yes`).
- **Strict Double-Entry Invariants**: Every confirmed entry produces balanced Debits and Credits with source provenance (`source = 'whatsapp_confirmed'`).
- **Instant Real-Time UI Sync**: As soon as a WhatsApp entry is confirmed, the web dashboard recalculates P&L, updates operating expenses, and highlights the entry in real time via Server-Sent Events (SSE).

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2. High-Level Architecture & Data Flow



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3. Meta WhatsApp Cloud API Configuration

To operate the live WhatsApp assistant, configure credentials in your `.env` file and set up the webhook in Meta for Developers.

Required Environment Variables (`.env`)

```
# ═══════════════════════════════════════════════════════════════════════
# WHATSAPP CLOUD API CONFIGURATION (META FOR DEVELOPERS)
# ═══════════════════════════════════════════════════════════════════════
WHATSAPP_TOKEN=EAAG...                    # Permanent System User Access Token
WHATSAPP_PHONE_NUMBER_ID=105938472910     # Assigned Phone Number ID from Meta
WHATSAPP_BUSINESS_ACCOUNT_ID=waba_10293   # WhatsApp Business Account ID
WHATSAPP_VERIFY_TOKEN=gross_verify_token # Secret string you choose for webhook handshake
WHATSAPP_APP_SECRET=a8b3c4d5e6f7...       # App Secret for HMAC-SHA256 signature verification

# Optional: Claude AI for fallback parsing (built-in regex runs at $0 cost)
ANTHROPIC_API_KEY=
```

Step-by-Step Meta Setup Guide

Step 1: Create a Meta App

1. Go to Meta for Developers and log in with your Facebook account.
2. Click **My Apps** -> **Create App**.
3. Select **Other** as the use case -> Select **Business** as the app type.
4. Name your app (e.g. `Nirvul360 Assistant`) and link your Meta Business Account.

Step 2: Add WhatsApp Product

1. In the App Dashboard, scroll to **Add Products to your app** and click **Set Up** on **WhatsApp**.
2. You will be redirected to the **API Setup** page.
3. Meta will assign you a temporary test phone number and a test **Phone number ID**.
4. For production, click **Add Phone Number** to register your official business SIM/landline.

Step 3: Generate Permanent System User Access Token

> 📌 **IMPORTANT**
> The temporary token on the API Setup page expires after 24 hours. For production 24/7 reliability, create a Permanent System User Token:

1. Open Meta Business Manager.
2. Go to **Users** -> **System Users** -> Click **Add**.

3. Set role to **Admin**.
4. Click **Generate New Token**, select your WhatsApp App, and grant these permissions: - `whatsapp_business_messaging` - `whatsapp_business_management`
5. Copy the generated token into `WHATSAPP_TOKEN` in your `.env`.

Step 4: Configure Webhook

1. In Meta App Dashboard, navigate to **WhatsApp** -> **Configuration**.
2. Under **Webhook**, click **Edit**: - **Callback URL**: `https://<your-public-domain>/webhooks/whatsapp` - **Verify Token**: Enter the exact same string set in `WHATSAPP_VERIFY_TOKEN` (e.g. `gross_verify_token`).
3. Click **Verify and Save**. Meta will make an instant `GET` request to your backend with `hub.challenge`.
4. Click **Manage Webhook Fields**, find `messages`, and check **Subscribe**.

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4. Natural Language & Command Engine

The assistant uses an optimized, zero-latency natural language parser in `backend/app/whatsapp/parser.py`. It detects digits in both English and Bengali, maps financial heads, and identifies operational intents.

A. Operational & Miscellaneous Expenses

Command / Message	Detected Head	Canonical Account	Action on "হ্যাঁ" Confirmation
`packaging 500 tk` / প্যাকেজিং ৫০০ টাকা	`packaging`	`5200` Packaging Expense	Debits Packaging Expense ৳500, Credits Cash ৳500
`courier charge 150 tk` / কুরিয়ার চার্জ ১৫০ টাকা	`courier_delivery`	`5150` Courier Delivery Expense	Debits Courier Expense ৳150, Credits Cash ৳150
`staff salary 12000 tk` / বেতন ১২০০০ টাকা	`salary`	`5250` Salaries & Wages	Debits Salary Expense ৳12,000, Credits Cash ৳12,000
`shop rent 15000 tk` / দোকান ভাড়া ১৫০০০ টাকা	`rent`	`5300` Rent Expense	Debits Rent Expense ৳15,000, Credits Cash ৳15,000
`electric bill 2400 tk` / কারেন্ট বিল ২৪০০ টাকা	`utilities`	`5350` Utilities Expense	Debits Utilities Expense ৳2,400, Credits Cash ৳2,400
`rickshaw fare 80 tk` / যাতায়াত খরচ ৮০ টাকা	`transport`	`5400` Travel & Conveyance	Debits Transport Expense ৳80, Credits Cash ৳80
`facebook boost 2500 tk` / বিজ্ঞাপন ২৫০০ টাকা	`ad_spend`	`5050` Advertising & Marketing	Debits Ad Spend ৳2,500, Credits Cash ৳2,500
`nasta 120 tk` / নাস্তা খরচ ১২০ টাকা	`other_expense`	`5500` Miscellaneous Expense	Debits Misc Expense ৳120, Credits Cash ৳120; Flags unallocated head

B. Owner Withdrawals vs. Operating Expenses

> **CAUTION**
> Double-entry integrity requires that owner personal cash withdrawals are **never** logged as operating expenses, as doing so would falsely depress company Net Profit and distort tax accounting.

- **Message**: `tulilam 5000 tk` / ব্যক্তিগত উত্তোলন ৫০০০ টাকা / `withdraw 5000`
- **Detected Intent**: `withdrawal`
- **Accounting Action**: Debits `3010 Owner Drawings (Equity)`, Credits `1010 Cash on Hand`. Operating expense remains unchanged; net profit is preserved.

C. Order & Consignment Updates

- **Message**: `order #1894 delivered` / `অর্ডার ১০৫ ডেলিভারি সম্পন্ন`
- **Detected Intent**: `order_status_update`
- **Action**: Transitions order state in `orders` table to `delivered`, triggers COGS recognition, updates real-time pipeline on dashboard.

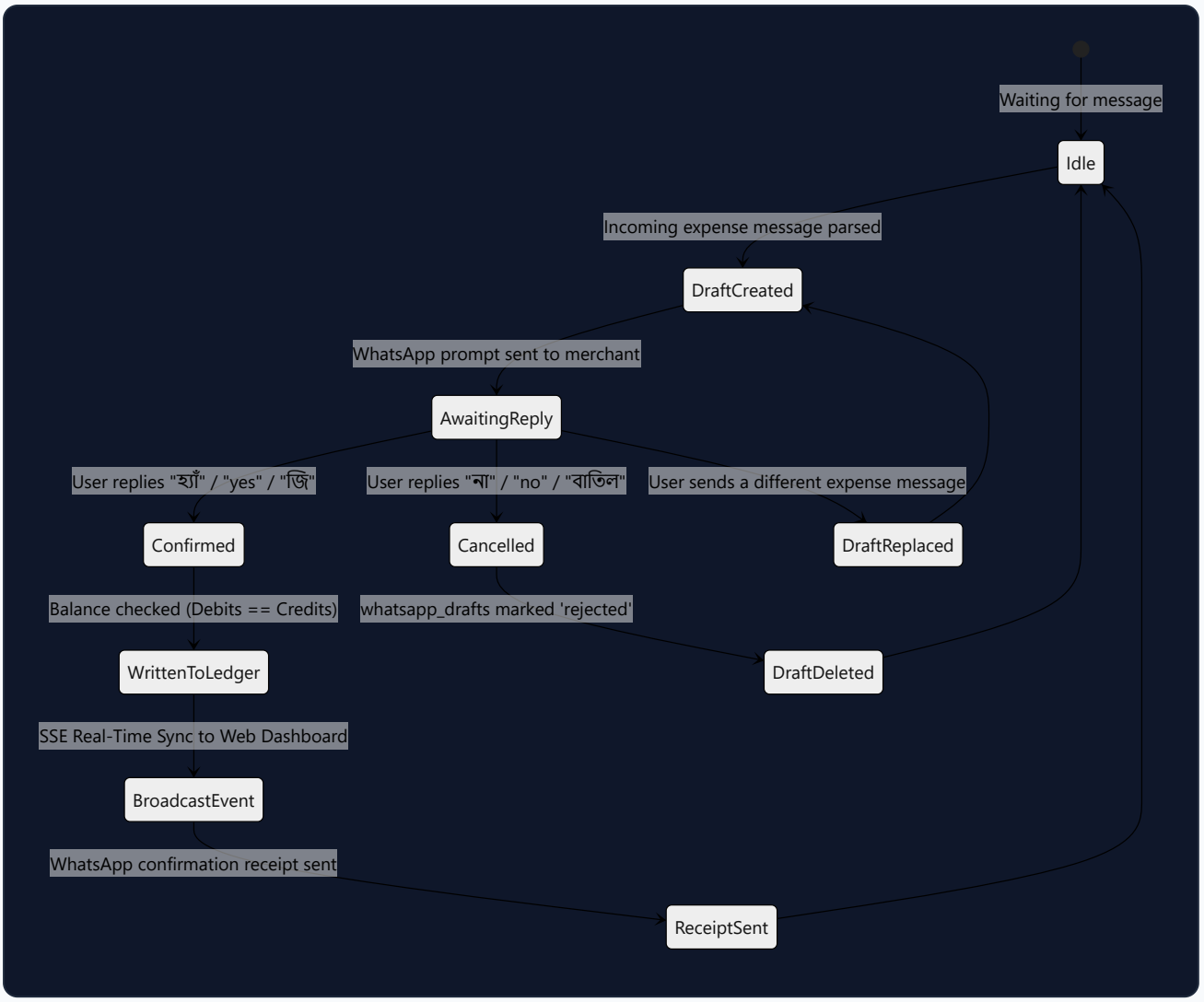
D. Real-Time Performance & P&L Queries

- **Message**: `আজকের হিসাব`, `today's profit`, `summary`, `লাভ`
- **Action**: Queries live P&L engine (`reports.py`) and replies immediately: `text` 📊 `আজকের সংক্ষিপ্ত হিসাব: • ডেলিভারি সেলস: ৳১৮,৪৫০.০০ • পণ্যের ক্রয়মূল্য (COGS): ৳৯,২০০.০০ • আজকের নিট লাভ: ৳৫,১৩০.০০`

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5. 2-Step Confirmation State Machine

To prevent erroneous posts or misparsed text from contaminating financial statements, the assistant implements a rigid **2-Step Confirmation State Machine**:



State Machine Rules

1. **Initial Message**: - Creates a record in `whatsapp_drafts` with `status = 'pending'`, `stage = 'confirm'`. - The double-entry ledger is **100% untouched**. - Replies to user: ```text 📝 এই এন্ট্রিটি ড্রাফট করা হয়েছে: খাত: প্যাকেজিং ও ব্যাগ পরিমাণ: ৳৫০০.০০ তারিখ: 2026-11-11

   নিশ্চিত করতে 'হ্যাঁ' লিখুন, অথবা বাতিল করতে 'না' লিখুন। `2. **Confirmation (`is_yes`)**: - Recognized words: `হ্যাঁ`, `হাঁ`, `জি`, `ঠিক আছে`, `yes`, `y`, `ok`, `okay`, `thik ase`. - Transitions draft `status = 'confirmed'`. - Calls `record_expense()` inside an ACID database transaction. - Generates double-entry journal entry with `source = 'whatsapp_confirmed'`. - Broadcasts real-time event via SSE to connected browsers. - Sends confirmation receipt:` text ✅ লিখে রাখা হয়েছে (জার্নাল #104)। ড্যাশবোর্ডে 'WhatsApp' ট্যাগসহ দেখা যাবে। `` 3. **Cancellation (`is_no`)**: - Recognized words: `না`, `ভুল`, `cancel`, `বাতিল`, `থাক`, `no`, `n`. - Transitions draft `status = 'rejected'`. - Sends feedback:` ❌ এন্ট্রি বাতিল করা হয়েছে। কোনো তথ্য লেজারে লেখা হয়নি।`

6. Double-Entry Accounting Ledger Integration

Nirvul360 adheres to strict double-entry bookkeeping rules. Every WhatsApp transaction satisfies the accounting equation: $$\Delta \text{Assets} = \Delta \text{Liabilities} + \Delta \text{Equity} + (\text{Revenues} - \text{Expenses})$$

Journal Entry Posting (`source = 'whatsapp_confirmed'`)

When a ৳500 packaging expense is confirmed:

```
-- 1. Insert Master Journal Entry
INSERT INTO journals (merchant_id, entry_date, description, source, reference, created_at)
VALUES (1, '2026-11-11', 'WhatsApp: packaging 500 tk', 'whatsapp_confirmed', 'draft_42', '2026-11-11

-- 2. Insert Debit Line (Packaging Expense 5200)
INSERT INTO journal_lines (journal_id, account_id, debit_poisha, credit_poisha)
VALUES (104, (SELECT id FROM accounts WHERE code = '5200'), 50000, 0);

-- 3. Insert Credit Line (Cash on Hand 1010)
INSERT INTO journal_lines (journal_id, account_id, debit_poisha, credit_poisha)
VALUES (104, (SELECT id FROM accounts WHERE code = '1010'), 0, 50000);
```

Invariant Checks

1. `SUM(debit_poisha) == SUM(credit_poisha)` enforced by `assert_balanced()` in `backend/app/domain/ledger.py`.
2. Provenance tag `whatsapp_confirmed` is permanently stamped on the journal record for financial auditability.

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7. Real-Time Dashboard & UI Synchronization

As soon as a draft is committed: 1. **Cache Invalidation**: - Calls `invalidate_dashboard_cache()` and `invalidate_expenses_cache()`. 2. **Server-Sent Events (SSE)**: - Backend event bus broadcasts: `json { "event": "whatsapp_confirmed", "data": { "ref": "জার্নাল #104", "intent": "expense", "merchantId": 1 } }` 3. **Frontend Dynamic Reaction**: - `frontend/src/components/ExpensesView.tsx` : - Increments the WhatsApp Confirmed badge count. - Refreshes the 6 expense category cards. - Prepends the newly booked entry to the **Recent entries** table with the green pill: `WhatsApp · confirmed just now`. - **Executive Dashboard**: - Operating expenses line item increases. - Net profit and Net margin recalculate instantly.

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8. Interactive In-App Simulator (Zero-API Key Testing)

You do **not** need a live Meta WhatsApp developer account to test or demonstrate this feature. The project includes an **In-App Simulator**:

- **Location in UI**: Open the WhatsApp Assistant drawer from the top navigation bar or the floating action button.
- **Component**: `frontend/src/components/WhatsAppAssistantView.tsx`
- **Backend Endpoints**:
- `POST /api/whatsapp/simulate`: Simulates an incoming text message as if sent by the merchant's registered phone.
- `POST /api/whatsapp/confirm`: Simulates tapping or typing "হ্যাঁ".
- `POST /api/whatsapp/reject`: Simulates tapping or typing "না".
- `GET /api/whatsapp/overview`: Returns telemetry, authorized staff phone numbers, and past 50 confirmation logs.

```
[Simulator Screen Example]
User:  packaging 500 tk
Bot:   📝 এই এন্ট্রিটি ড্রাফট করা হয়েছে:
         খাত: প্যাকেজিং ও ব্যাগ
         পরিমাণ: ৳৫০০.০০
       নিশ্চিত করতে 'হ্যাঁ' লিখুন, অথবা বাতিল করতে 'না' লিখুন।
User:  হ্যাঁ
Bot:   ✅ লিখে রাখা হয়েছে (জার্নাল #104)। ড্যাশবোর্ডে 'WhatsApp' ট্যাগসহ দেখা যাবে।
```

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9. Security, Authentication & Signature Verification

1. **HMAC-SHA256 Payload Signature Check**: Every incoming request to `POST /webhooks/whatsapp` is validated using the header `x-hub-signature-256` and `WHATSAPP_APP_SECRET`: `python expected = hmac.new(secret.encode('utf-8'), raw_body, hashlib.sha256).hexdigest() if not hmac.compare_digest(signature[7:], expected): raise HTTPException(status_code=401, detail="signature mismatch")`
2. **Phone Whitelist & Role-Based Authorization**: - Unregistered numbers receive: `দুঃখিত, এই নম্বরটি কোনো মার্চেন্ট অ্যাকাউন্টের সাথে যুক্ত নেই।` - Staff roles (`staff` , `packer`) are capped at operational expenses (e.g. up to ৳5,000). - Only `owner` and `admin` roles are permitted to record withdrawals (`উত্তোলন`).
3. **Idempotency & Replay Prevention**: - Meta message IDs (`wamid`) are deduplicated to ensure duplicate webhook deliveries do not create duplicate ledger entries.

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10. Codebase Reference & Directory Structure

Layer	File Path	Primary Function
Parser	`backend/app/whatsapp/parser.py`	NLP keyword detection, Bengali digit conversion, amount extraction.
State Machine	`backend/app/whatsapp/assistant.py`	2-step confirmation loop, draft creation, commit/reject handlers.
Meta Client	`backend/app/whatsapp/client.py`	Outbound WhatsApp message dispatcher via Meta Graph API v18.0.
Webhook Handler	`backend/app/api/webhooks.py`	Inbound endpoint for Meta challenge handshake & incoming messages.
API Endpoints	`backend/app/api/whatsapp.py`	Authenticated dashboard endpoints (`/overview` , `/conversation` , `/simulate`).
Double-Entry Engine	`backend/app/domain/entries.py`	Books balanced journals with `source = 'whatsapp_confirmed'` .
Frontend Simulator	`frontend/src/components/WhatsAppAssistantView.tsx`	Live interactive chat simulator and audit log viewer.
Expenses View	`frontend/src/components/ExpensesView.tsx`	Displays real-time expense head breakdown and WhatsApp-tagged entries.